
```
import matplotlib.pyplot as plt
xs = [x/100 for x in list(range(201))]
ys = [ball_trajectory(x) for x in xs]
plt.plot(xs,ys)
plt.title('The Trajectory of a Thrown Ball')
plt.xlabel('Horizontal Position of Ball')
plt.ylabel('Vertical Position of Ball')
plt.axhline(y = 0)
plt.show()
```

[Listing 1-2](#): Plotting a hypothetical ball trajectory between the moment it is thrown (at $x = 0$) and when it hits the ground again (at $x = 2$)

The output ([Figure 1-1](#)) is a nice plot that shows the path our hypothetical ball is expected to follow through space. This pretty curved path is similar for every moving projectile that's influenced by gravity and has been poetically called *Gravity's Rainbow* by the novelist Thomas Pynchon.

Not all balls will follow this exact path, but this is one possible path that a ball could follow. The ball starts at 0, and it goes up and then down exactly like we are used to seeing balls go up and down, from the left of our field of view to the right.